

```
def is_safe(board, row, col, n):
```

```
    # Check column
    for i in range(row):
        if board[i] == col:
            return False

    # Check left diagonal
    for i in range(row):
        if abs(board[i] - col) == abs(i - row):
            return False

    return True
```

```
def solve_n_queens(board, row, n):
```

```
    if row == n:
        return True

    for col in range(n):

        if is_safe(board, row, col, n):

            board[row] = col

            if solve_n_queens(board, row + 1, n):
                return True

            board[row] = -1

    return False
```

```
n = int(input("Enter number of queens: "))
```

```
board = [-1] * n
```

```
if solve_n_queens(board, 0, n):
```

```
    print("Solution:")

    for row in range(n):

        for col in range(n):

            if board[row] == col:
                print("Q", end=" ")
            else:
                print(".", end=" ")

        print()
```

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2024)
on win32
```

```
Enter "help" below or click "Help" above for more
```

```
>>>
```

```
==== RESTART: C:/Users/ROOPA SAI DIVYA/OneDrive
```

```
Enter number of queens: 8
```

```
Solution:
```

```
Q . . . . .
. . . . Q . .
. . . . . Q
. . . . . Q .
. . Q . . . .
. . . . . Q .
. Q . . . . .
. . Q . . . .
```

```
>>>
```